Tufts Dental Dataset - Exploration Report

Generated: 2026-03-27 05:05

1. File Inventory

Radiographs:	1000
Maxillomandibular:	1000
Teeth masks:	308
Expert masks:	439
Expert gaze:	1000
Student gaze:	941
Total radiographs:	1000

2. Radiograph Image Quality

Avg height:	840 px (840, 840)
Avg width:	1615 px (1615, 1615)
Mean Michelson contrast:	0.999

3. Annotation Summary

Expert records:	1000
Bbox records:	1000
Avg teeth per image:	?
FDI column found:	NOT FOUND

4. Expert Diagnosis Categories

Category	Count	%
other	1000	100.0%

5. Bone Loss — Subset Analysis

Total bone loss records:	0 (0.0%)

Severity	Count

6. Mask Analysis

Maxillomandibular pixel values:	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 1
Teeth mask pixel values:	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 1

Maxillomandibular masks cover full jaw — useful for jaw segmentation. Teeth masks are the direct input for the bone loss model (only 300 images). Pixel values must be verified: if values = tooth IDs (1-32), masks are instance segmentation. If values = class IDs, masks are semantic.

7. Dataset Coverage Matrix

OPG only:	1000
OPG + maxillo mask:	1000
OPG + teeth mask:	308
OPG + expert label:	0

OPG + bbox:	0
OPG + teeth mask + expert:	0
OPG + bbox + expert:	0
FULL (OPG+teeth+bbox+expert):	0

8. Pipeline Decision

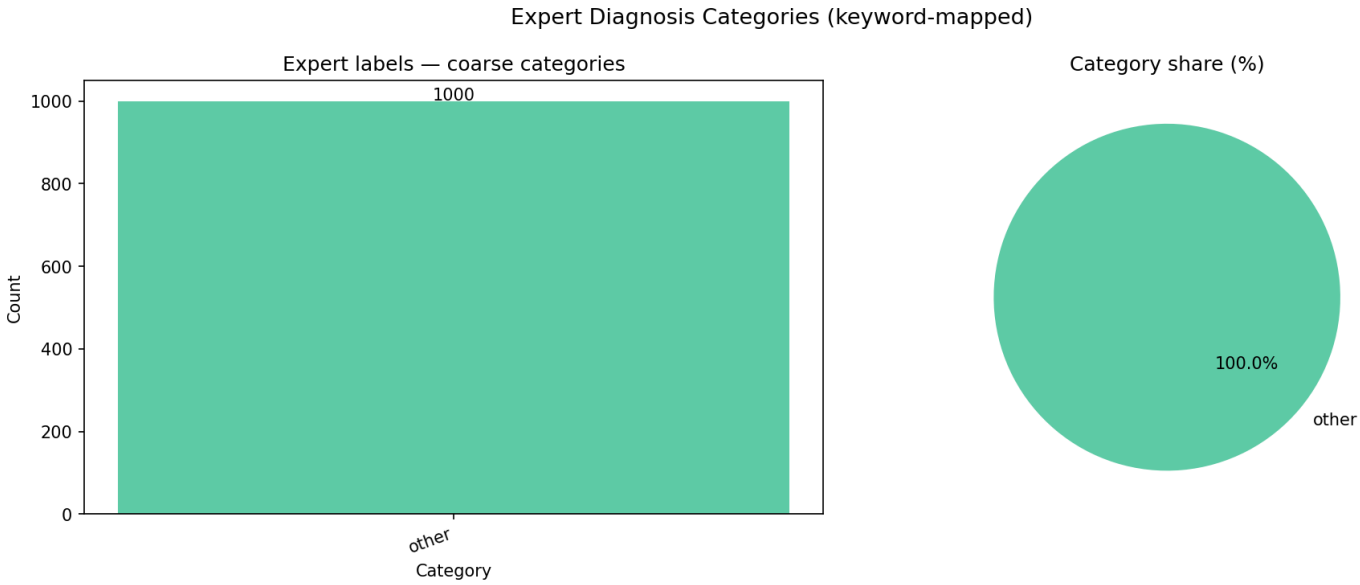
DECISION: Only 0 fully-annotated samples. Consider classification-only approach or data augmentation.

1. Confirm pixel value encoding in teeth_mask (tooth ID vs class ID).
2. Map expert free-text labels to per-tooth severity using bbox image_id.
3. Use bbox JSON to generate expanded crops (1.8x) for bone loss model.
4. Add +/-15% bbox jitter during training to simulate YOLO noise.
5. Stratified train/val split on the 300 teeth-mask images.
6. Set per-class weights: w_i = total / (n_classes x count_i).
7. Evaluation target: per-class F1 >= 0.70 for all severity levels.

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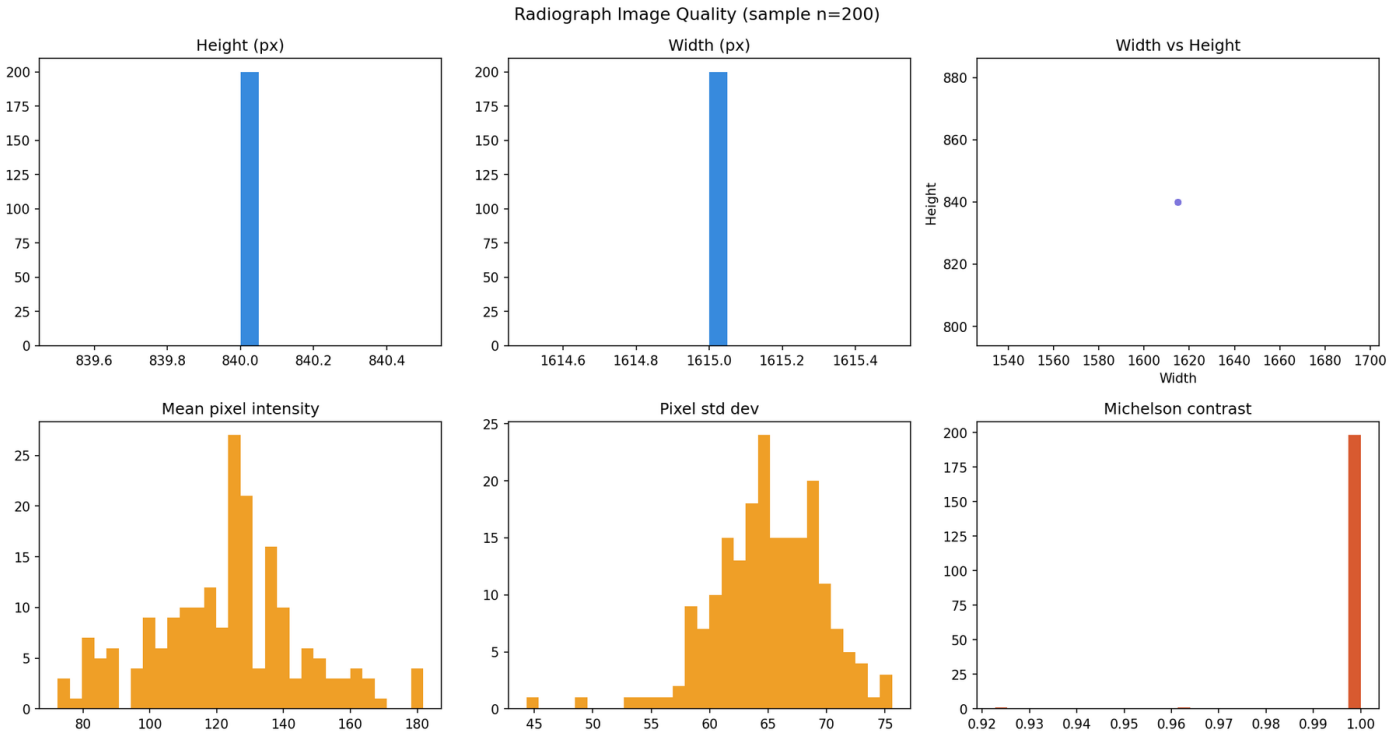
9.1 Expert Diagnosis Categories



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9.2 Radiograph Image Quality



9.3 Tooth Bounding Box Statistics

[Not generated]

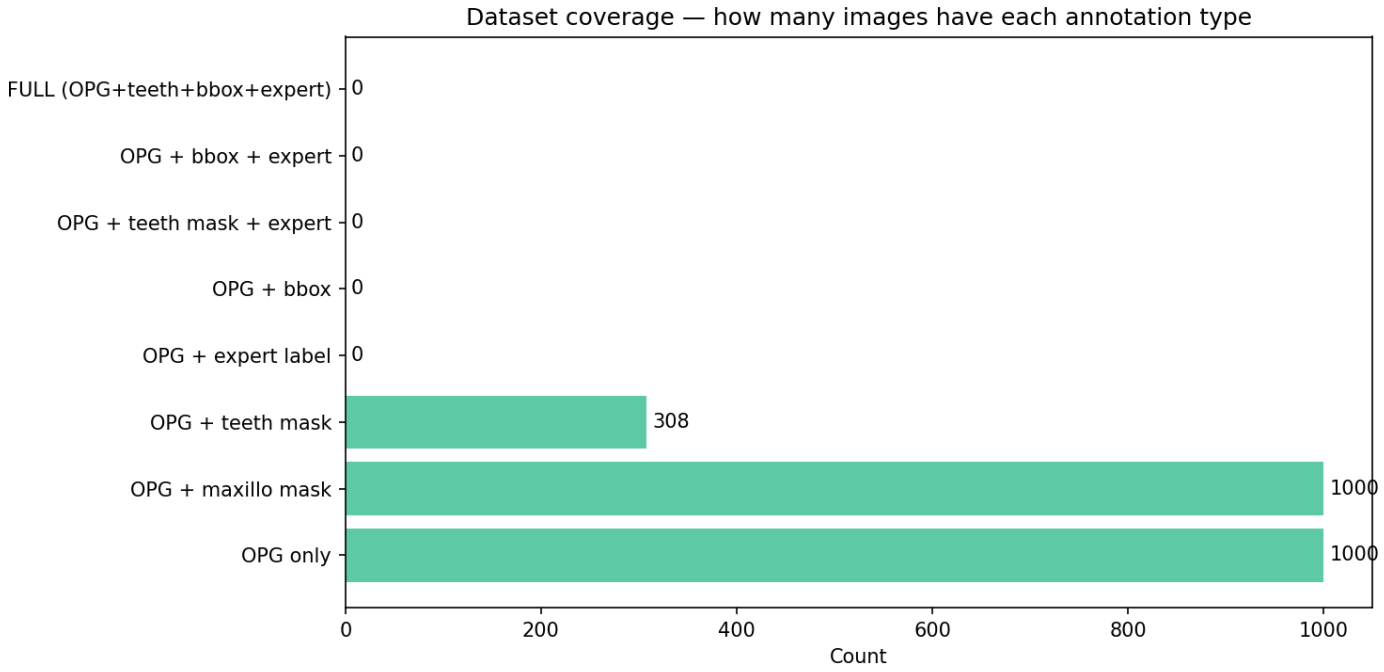
9.4 Bbox Overlaid on OPG

[Not generated]

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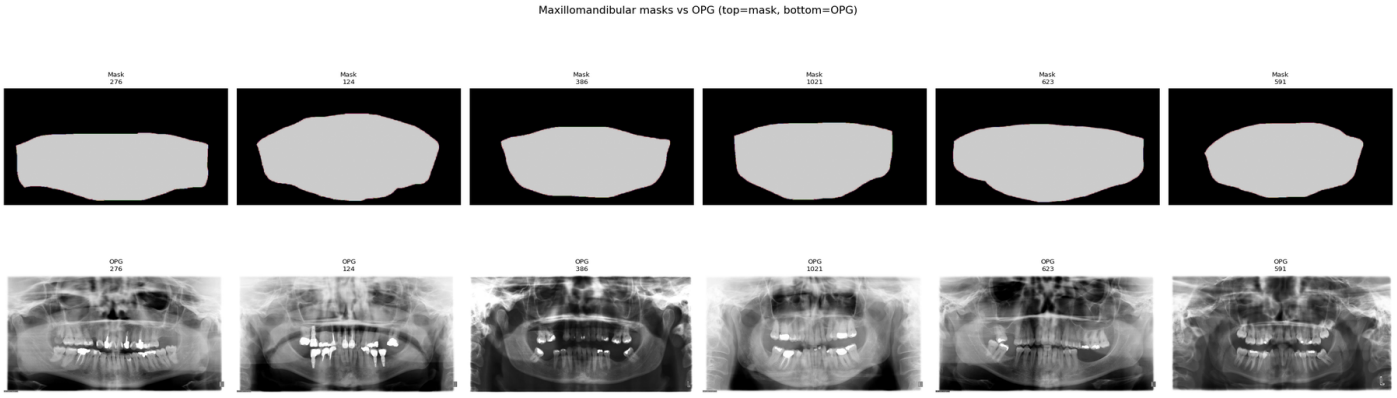
9.5 Dataset Coverage Matrix



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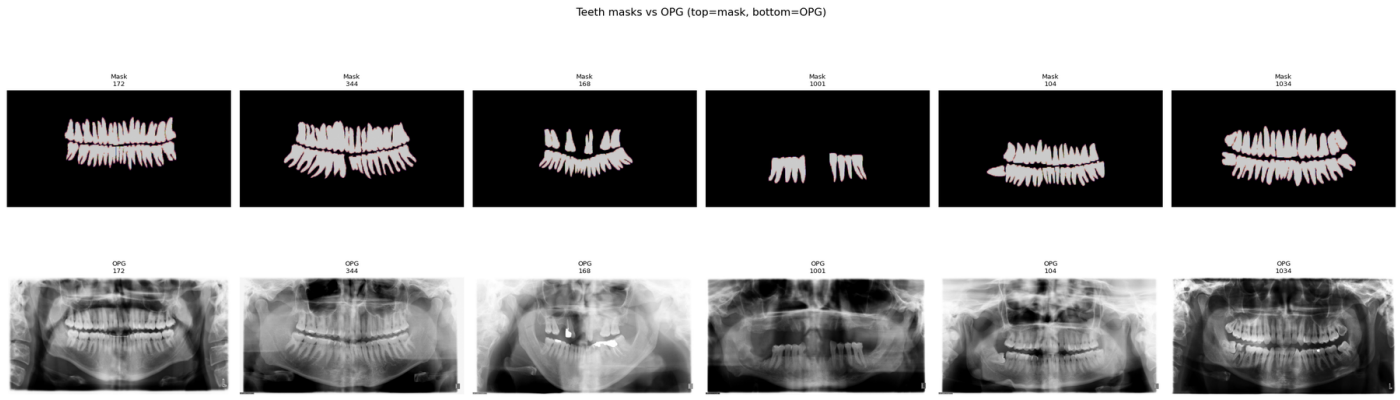
9.6 Maxillomandibular Masks vs OPG



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9.7 Teeth Masks vs OPG



9.8 Crop Simulation

[Not generated]

9.9 Tooth Label / FDI Coverage

[Not generated]